

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 28-12-2025**

Time: 02.00Pm to 05.00Pm **GTA-09** **Max. Marks: 180**

28-12-2025_Sr. XL-500, Sr.S60(N&S) & LIIT_Jee-Adv(2025-P2)_GTA-09_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2025-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 8)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 9 – 16)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 25 – 32)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 40)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 41 – 48)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Q is a point on the auxiliary circle of an ellipse. P is the corresponding point on ellipse. N is the foot of perpendicular from focus S, to the tangent of auxiliary circle at Q. Then

A) $SP = SN$ **B)** $SP = PQ$ **C)** $PN = SP$ **D)** $NQ = SP$

2. If N be the number of natural numbers less than 2020 which can be expressed in the form of $\left[x\left[x\right]\right]$ for some positive real x, then N is not divisible by

(Note: [.] denotes greatest integer function)

A) 3 **B)** 5 **C)** 7 **D)** 11

3. The domain of the function

$f\left(x\right)=\left[2\tan\pi x\right]^{\log_{[2\tan\pi x]}\left(g\left(x\right)\right)}$, where $g\left(x\right)=\left(\dfrac{x^2+2x-3}{4x^2-4x-3}\right)$ contain the interval

$\left[n+\dfrac{1}{4},n+\dfrac{1}{2}\right)$, where n is an integer. Then n can be equal to

(Note: [.] denotes greatest integer function)

A) -2 **B)** -1 **C)** 0 **D)** 1

4. If z is a complex number such that $\left|z\right|=1$, then the maximum value of

$\left|z^4+z^3-2z^2i+z+1\right|$ is

A) $\sqrt{29}$ **B)** $\sqrt{22}$ **C)** $\sqrt{28}$ **D)** $\sqrt{20}$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. If $\int 4(3-2x)^{-2}\left(\frac{3-2x}{3+2x}\right)^{\frac{1}{3}}dx=\frac{3}{\alpha}\left(\frac{3+2x}{3-2x}\right)^{\beta/\gamma}+C$ (β & γ are prime numbers), then select the CORRECT alternative/s

A) α,β,γ are in G.P. **B)** α,γ,β are in H.P.

C) $\alpha,2\beta,\gamma$ are in H.P. **D)** $\alpha=\beta\gamma$

6. If $A=\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}$, then which of the following is/are CORRECT?

A) $A^3-A^2=A-I$ **B)** $Det\left(A^{2010}-I\right)=0$

C) $A^{50}=\begin{bmatrix}1&0&0\\25&1&0\\25&0&1\end{bmatrix}$ **D)** $A^{50}=\begin{bmatrix}1&1&0\\25&1&0\\25&0&1\end{bmatrix}$

7. Let $f(x)=\begin{cases}\left[\frac{2\left(\tan^{-1}x-x\right)}{x^2}\right];x<0\\ \left(x^2-4\right)\left|x^2-2x\right|;x\geq 0\end{cases}$; then which of the following option(s) is/are correct?

([.] is G.I.F)

A) Number of points at which $f(x)$ is discontinuous in R is not finite

B) Number of points at which $f(x)$ is discontinuous in R is "0"

C) Number of points at which $f(x)$ is not differentiable in $(-1,\infty)$ is 1

D) Number of points at which $f(x)$ is not differentiable in R is "1"

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8. $f(x)=\int_{-x}^{x}\left[\tan^{-1}t+\tan^{-1}\frac{1}{t}\right]dt; x\in R-\{0\}$,where [.] denotes G.I.F, then which of the following option(s) is/are incorrect?

A) $f(x)$ is an even function

B) $f(x)$ is an odd function

C) The number of real solution of $f(x)=|x|+|x+1|-1$ is 1

D) $f(4)=8$

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

9. Let $A=\left\{x\middle|x^3+x^2-px+q=0, p,q\in R\right\}$ and $B=\left\{x\middle|x^2-qx+2=0, q\in R\right\}$ be the sets . If $n(A\cap B)=2$ *and* $x_0\in A-B$ then the value of $|p-q+x_0|$ is equal to …..($n(A)$ means number of elements in the set A)

10. Consider the function $f(x)=2^{2022a}x^3+3^{2020}x^2-\left(2^{2021a+1}+2019\right)x+2018$.If k_1 *and* k_2 are respectively the local maximum and local ,minimum values of $f(x)$ and two sets A and B are given as $A=\left\{x: f(x)=k_1 \text{ and } f'(x)\neq 0\right\}$ and $B=\left\{x: f(x)=k_2 \text{ and } f'(x)\neq 0\right\}$,then the sum of all elements of $A\cup B$ is $g(a)\forall a$, wherever it is defined. Then the number of positive integral divisor(s) of $g'\left(\frac{1}{1011}\right).\log_8 e$ *is* _____

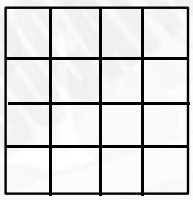
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11. In a plane rectangular coordinate system ,suppose there exists a point (a,b) on the circle with center coordinates $(r+1,0)$ and radius r satisfying $b^2 \geq 4a$ .Then the minimum value of r is _____

12. In a plane rectangular coordinate system , $\vec{e}$ is a unit vector ,vector $\vec{a}$ satisfies $\vec{a}.\vec{e}=2$, and $|\vec{a}|^2 \leq 5|\vec{a}+t\vec{e}|$ holds for any real number t. Then the range of $|\vec{a}|$ is $[\alpha,\beta]$, then the value of $\alpha+\beta$ is ___

13. Four letters, two "a"s and two "b"s , are filled into 16 cells of a matrix as shown in figure. It is required that each cell contains at most one letter, and each row or column cannot contain the same letters. Then the number of different ways that the matrix can be filled is ___



14. Number of ordered pairs (x,y) with $0 \leq x \leq \pi$ & $\pi \leq y \leq 2\pi$ satisfying the equation $(\sin x+\sqrt{3}\cos x)(\sin(x+y)+\sin(x-y))=4$ is

15. OA, OB, OC are the sides of a rectangular parallelopiped whose diagonals are OO', AA', BB' and CC'. D is the centre of rectangle $AC'O'B'$ and D' is centre of rectangle $O'A'CB'$. If sides OA, OB, OC are in ratio 1: 2: 3 and $\angle DOD'=\alpha$, then the value of $\frac{697\cos^2\alpha}{128}$ is ___ (Centre of rectangle is mid point of diagonal)

16. Given $f(z)=z^2-19z$, there are complex numbers z with the property that $z, f(z), and\ f(f(z))$ are the vertices of a right triangle in the complex plane with a right angle at $f(z)$. There are positive integers m and n such that one such value of z is $m+\sqrt{n}+11i$. Then the value of $m+n$ is…..

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PHYSICS **Max Marks: 60**

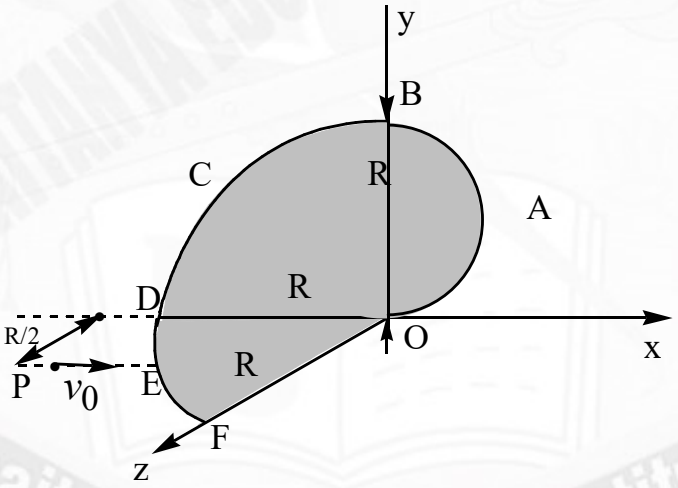
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

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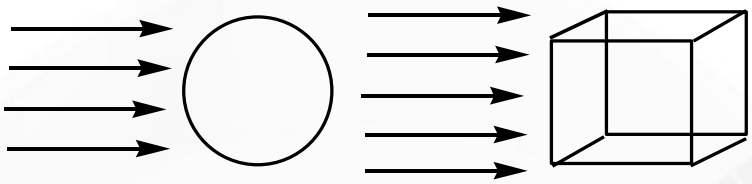
17. A folded plate OABCDEFO made of materials such that part OABO (say part (i)), BCDOB (say part (ii)) and DEFOD (say part (iii)) has mass 4m, m and m respectively. Part (i) is a uniform semicircular plate of radius R/2 and is on the xy plane. Part (ii) and (iii) each is a uniform quarter circular plate of radius R on system is free to rotate about y-axis. A particle P of mass m moving with velocity v_0 hits to a point located at the circumference of the part (iii) and sticks to it. The point is at a distance R/2 from x-axis as shown in the figure. Angular velocity of the combined system just after the collision is



A) $\frac{v_0}{4R}$ **B)** $\frac{2v_0}{5R}$ **C)** $\frac{3v_0}{5R}$ **D)** $\frac{v_0}{2R}$

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18. A parallel beam of radiation, is incident first on a highly conducting sphere and then on a highly conducting cube as shown. If both sphere and cube are made from same material (same emissivity) then which of the following is CORRECT :



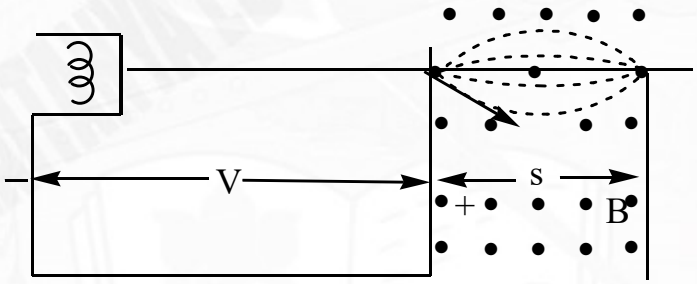
A) Steady state temperatures of sphere and cube are same

B) Steady state temperature of sphere will depend on its size

C) Steady state temperature of cube depends on tis dimensions

D) Steady state temperature of sphere is more than steady state temperature of cube

19. Figure shown an electron gun that emits electrons at very small angel with axis. The paraxial electrons emitted from the cathode (filament) are refocused at a distance s after accelerating through a potential difference V, s is given by (Take $\pi^2 = 10$):



A) $s = 2\sqrt{\dfrac{20mV}{eB^2}}$ **B)** $s = \sqrt{\dfrac{20mV}{eB^2}}$ **C)** $s = \sqrt{\dfrac{10mV}{eB^2}}$ **D)** $s = \sqrt{\dfrac{5mV}{eB^2}}$

20. Figure (a) shows plot of voltage across the capacitor as a function of the driving frequency for a sinusoidally driven electromagnetic oscillator LCR circuit. Figure (b) shows phase angle ϕ (phase difference between voltage and current) vs ω / ω_0 graph for same circuit, for three different quality factor graph 1, 2, 3 of figure (a) and each one can be matched by one of graphs a, b, c also of figure (b):

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A) Graph (3) corresponds to graph (a)

B) Graph (1) corresponds to graph (c)

C) The circuit of graph 1 has high quality factor

D) The circuit of graph 3 has high quality factor

figure (a)

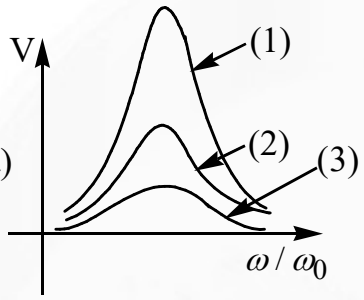
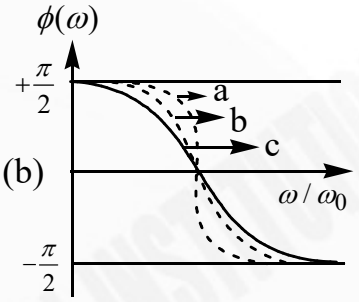


figure (b)



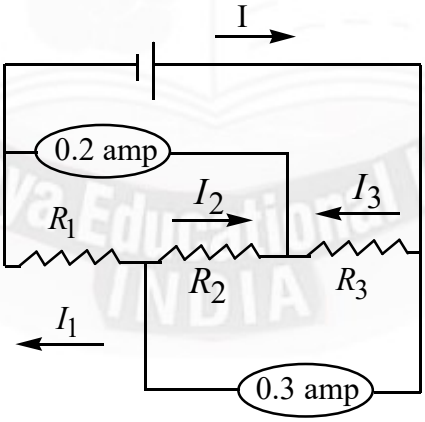
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions

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21. A circuit contains an ideal battery, three resistors and two ideal ammeters. The ammeters read 0.2 amp and 0.3 amp. After two of the resistor are switched (interchanged) the readings of ammeter do not change. Then



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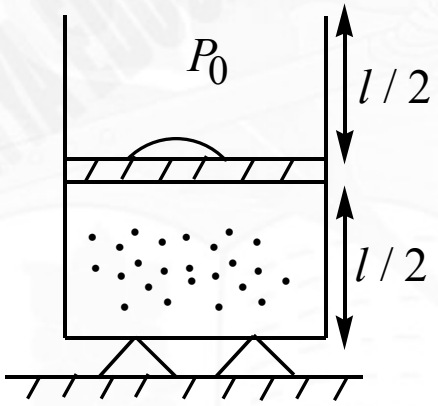
A) If $R_2 = R_1$, $I_1 = I_2 = 0.15\,amp$ and $I_3 = 0.05\,amp$.

B) If $R_2 = R_1$, the battery current is, I = 0.20 amp.

C) If $R_2 = R_3$, $I_2 = I_3 = 0.10\ amp$ and $I_1 = 0.20\ amp$.

D) If $R_2 = R_3$, the battery current is, I = 0.40 amp.

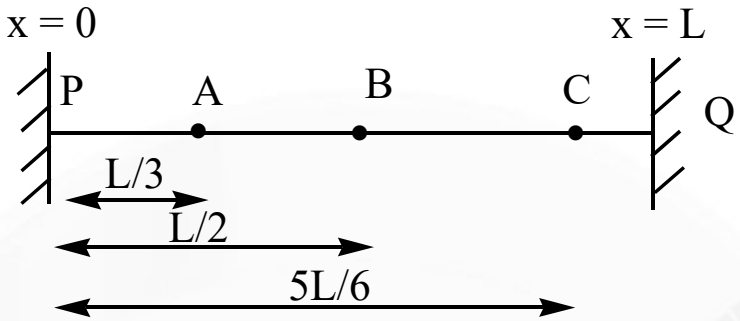
22. Figure shows initial state of an ideal gas trapped in a container with conducting walls and a piston (mass m) which can move without any friction. The container is placed on point supports and its walls are conducting. Assuming that atmospheric pressure is P_0 and the mass of the ideal gas is negligible as compared to the mass of the piston and the mass of container. Take the cross section area of piston to be A. The piston is slowly lifted by an external agent and held in its position. Let M be the maximum mass of container so that it may "lift off" while pulling the piston upwards and P_i be the pressure of ideal gas in initial state (i.e before piston is lifted and piston is at equilibrium). Pick the correct choice(s):



A) $P_i = P_0 + \frac{mg}{A}$ **B)** $P_i = \frac{P_0}{2} + \frac{mg}{2A}$ **C)** $M = \frac{m}{2}$ **D)** $M = \frac{P_0 A}{2g} - \frac{m}{2}$

23. Standing waves are established on a string of length L such that A is a node and B is an immediate anti node. Oscillation amplitude for point B is a_0. Let a' be the oscillation amplitude for point C. Pick the suitable option(s) for correct value of a' and the possible equation(s) for standing waves on the string.

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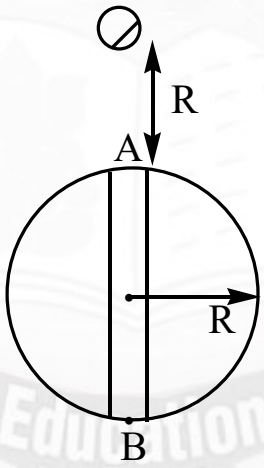
A) $a' = \frac{a_0}{2}$

B) $y = a_0 \sin\left(\frac{6\pi x}{L}\right)\cos(\omega t)$

C) $a' = a_0$

D) $y = a_0 \sin\left(\frac{3\pi x}{L}\right)\sin(\omega t)$

24. A small particle is dropped from a height R in front of a narrow tunnel dug inside the earth (along a diameter). Let M be the mass of earth, R be radius of earth. Let v_0, T be speed of particle when it reaches A and time taken by particle to go from A to B respectively. Assuming mass of particle to be negligible as compare to mass of earth, pick the correct option(s).



A) $V_0 = \sqrt{\frac{GM}{2R}}$

B) $V_0 = \sqrt{\frac{GM}{R}}$

C) $T = \frac{\pi}{2}\sqrt{\frac{R^3}{GM}}$

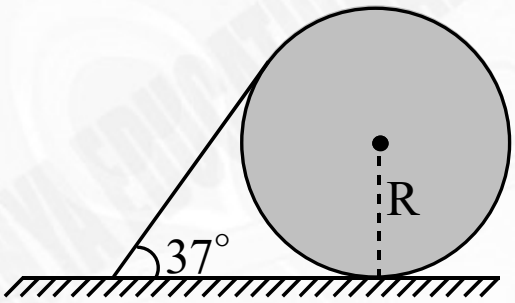
D) $T = \pi\sqrt{\frac{R^3}{GM}}$

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SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
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- **Zero Marks:0** in all other cases

25. A stick of mass density $\lambda = 8kg/m$ rests on a disc of radius R = 20 cm as shown in the figure. The stick makes an angle $\theta = 37^\circ$ with the horizontal and is tangent to the disc at its upper end. Friction exists at all points of contact and assume that it is large enough to keep the system at rest. Find the friction force (in Newton) between the ground and the disc. (take $g = 10m/s^2$)

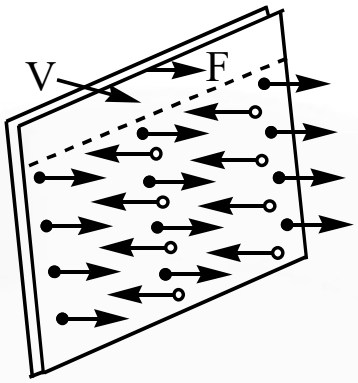


26. A point source S emits unpolarized light uniformly in all directions .At two points A and B, the ratio $r = \frac{I_A}{I_B}$ of the intensities of light is 2. If a set of two polaroids having 45^0 angle between their pass-axes is placed just before point B, then the new value of r will be _____.

27. A thin heavy metal plate is being bombarded by a perpendicular beam of gas particles from both sides as shown in the figure. The solid dots are representing the molecules hitting from left side and the faint dots are the molecules hitting from right side. The mass of these gas particles is $m = 10^{-26}kg$ and velocity before hitting is $v_0 = 5m/s$. Volume density of the gas particles on both sides is $n = 10^{25}$ per m^3. Each beam has an area $A = 1m^2$ and the collisions are perfectly elastic. What is the external force F (in Newton) required to move the plate with a constant velocity v = 2 m/s.

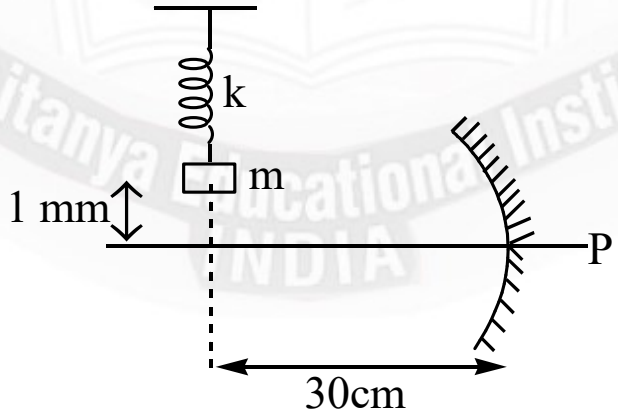
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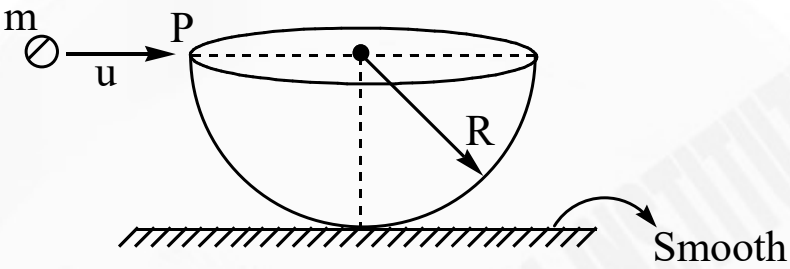
28. One possible method for revealing the presence of concealed nuclear weapons is to detect the neutrons emitted in the spontaneous fission of ^{240}Pu in the warhead. In an actual trial, a neutron detector of radius 1.0m carried on helicopter measured a neutron flux of $50s^{-1}$ at a distance of 100m from a missile warhead, that emits like a point source . Estimate the mass of ^{240}Pu (in kg) in the warhead .The mean life for spontaneous fission in is ^{240}Pu is 6×10^{18} sec and 2.5 neutrons on the average are emitted in each fission (Avogadro's number $=6\times10^{23}$ *per mole*)

29. Figure shows a small block suspended vertically using an ideal spring. Initially spring is in natural length and the mass is 1 mm above the principal axis of a concave mirror whose focal length is 20 cm. Given that mass m = 10 grams, k = 100 N/m, $g=10m/s^2$; the amplitude of the image of the block formed by concave mirror is A millimetres, find the integer value 'A'.



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30. A hollow hemispherical bowl of mass m, radius R is placed on a smooth horizontal surface. A particle (also of mass m) strikes the hemi spherical bowl at point P and sticks to it. The velocity of particle just after the collision in vertically upwards direction is $\frac{ku}{25}$; where k is an integer. Find k.



31. Consider the following nuclear reaction

$${}^2_1D + {}^2_1D \longrightarrow {}^3_1T + {}^1_1P$$

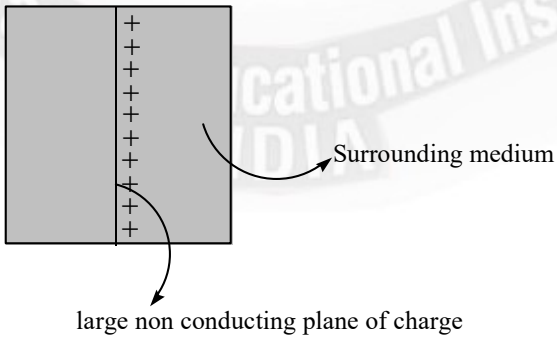
Given that $m\left({}^2_1D\right) = 2.01458$ amu,

$m\left({}^3_1T\right) = 3.01605$ amu,

$m\left({}^1_1P\right) = 1.00728$ amu.

The mass of deuterium $\left({}^2_1D\right)$ required per day in order to produce a power output of $10^9 W$ (with 50% efficiency) is $(n+0.3)kg$. where n is a digit. Find n.

32. A very large non-conducting plane carrying charge density σ uniformly distributed over it is placed in a medium with resistivity ρ and permittivity ε. The plane leaks charge through the surrounding medium and the charge density becomes $\frac{1}{n}$ times the initial value after a time $t = 6\rho\,\varepsilon\,\ell n(2)$ elapses. Find the integer value n. (Neglect the end effects).



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CHEMISTRY **Max Marks: 60**

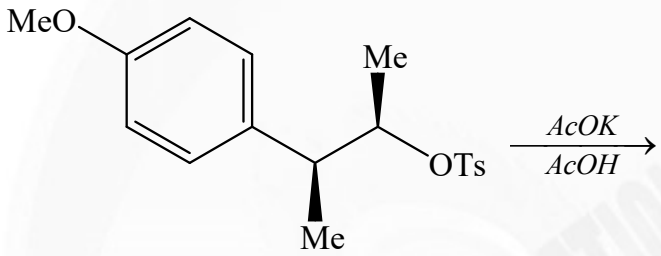
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

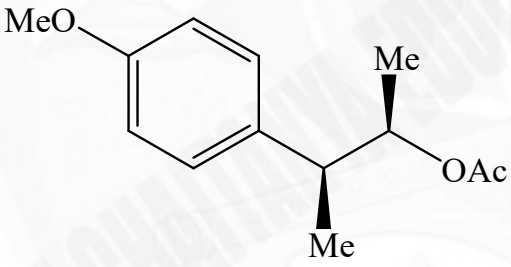
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

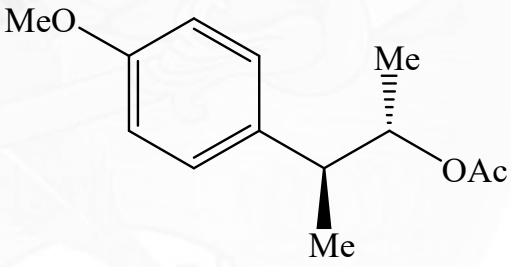
33. Solvolysis of the optically active compound X gives, mainly



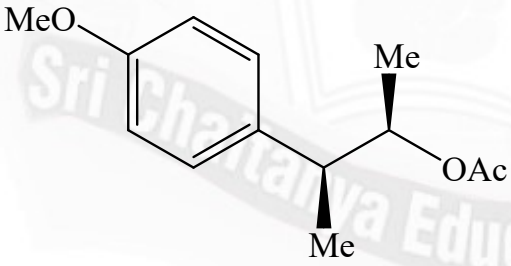
X



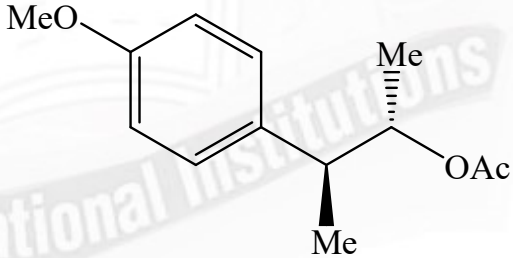
A) (Optically active)



B) (Optically active)



C) racemic (Optically inactive)



D) racemic (Optically inactive)

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34. Calculate pH of solution obtained by mixing two weak acid 100 ml 0.2M HA $(Ka = 3\times10^{-5})$ & 100 ml 0.2M HB $(Ka = 10^{-5})$.

A) $3-\frac{3}{2}\log 2$ **B)** $3-\log 3$ **C)** $3-\log 2$ **D)** $3-3\log 2$

35. Which of the following statements is correct regarding lanthanoids/lanthanoid ions? (lanthanoids: $_{58}Ce$ *to* $_{71}Lu$)

A) Europium $\left(_{63}Eu\right)$ has highest metallic radius.

B) Lutetium ion (Lu^{+3}) has highest ionic radius

C) They form hydroxides $M(OH)_3$, which are always hydrated oxides like $Fe_2O_3.xH_2O$, $Al_2O_3.xH_2O$

D) The E^0 value for the Ce^{+4}/Ce^{+3} couple is '$-1.74V$'.

36. Which of the following does not gives yellow coloured solution or precipitate?

A) $Ba^{+2} \xrightarrow{CrO_4^{2-}}$ **B)** $Fe^{+2} \xrightarrow{excess\ KCN}$

C) $Ag^{+} \xrightarrow{CrO_4^{2-}}$ **D)** $NH_4^{+} \xrightarrow{H_2[PtCl_6]}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

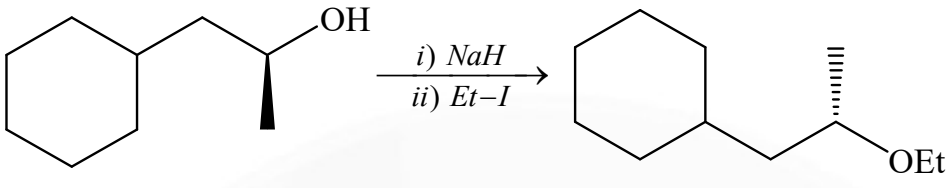
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

37. Which of the following is/are showing correct major product?

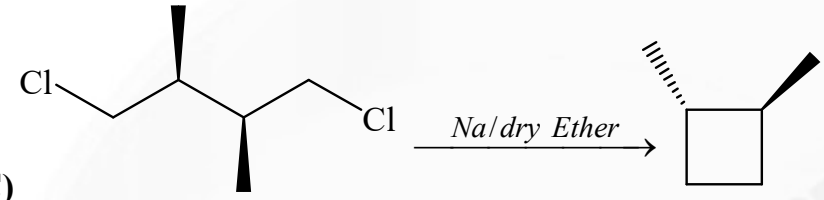
A) $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3 \xrightarrow[ii)\ CH_3-OH]{i)\ NaH} CH_3-\underset{\displaystyle O-CH_3}{\underset{|}{CH}}-CH_3$

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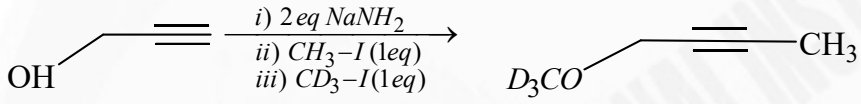
B)



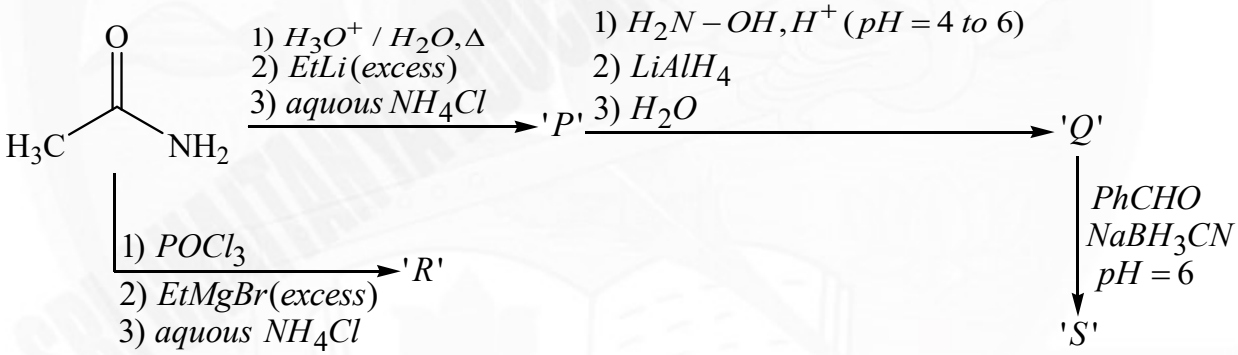
C)



D)



38. Which of the following is/are correct?



A) P and R are identical compounds.

B) Both P and R cannot give iodoform test.

C) Q can give carbylamines test.

D) Compound S can give Lieberman's nitroso test.

39. For the decomposition of $NH_3(g)$ at standard state.

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$

[Assume ΔH^0 and ΔS^0 of reaction remain constant; 'T' represents temperature in Kelvin scale & K_p represents equilibrium constant.]

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Choose the correct statement(s)

A) Graph between $\ln K_p vs\ 1/T$ is a straight line with positive slope and positive intercept.

B) Graph between $\ln K_p vs\ 1/T$ is a straight line with negative slope and positive intercept.

C) K_p increases with increase in temperature

D) At very low temperature reaction is non-spontaneous.

40. Which of the following statement(s) are correct regarding the oxides of nitrogen?

A) 'N-N' bond length in N_2O_3 is more than that in N_2O_4.

B) Both N_2O_3 and N_2O_4 are non-planar molecular

C) ONO bond angle in NO_2 is 134^0.

D) Covalence and oxidation state of nitrogen in N_2O_5 are 4 and +5 respectively

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +4** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

41. A tetrapeptide, made up of natural amino acids, has alanine as the N-terminal residue which is coupled to a chiral amino acid. Upon complete hydrolysis, the tetrapeptide gives glycine, alanine, phenylalanine and leucine. The number of possible sequences of the tetrapeptide is

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42.

$$\text{Butan-2-ol} \xrightarrow[\text{iii) Alc.KOH},\Delta]{\text{i) Pyridinium chlorochromate}\ \text{ii) AcOH/Cl}_2} \text{'P'}$$

$$\text{Methyl acetoacetate} \xrightarrow[\text{ii) aq } NH_4Cl\ \text{iii) } NaBH_4]{\text{i) NaOMe in MeOH, P}} \text{'R' (Major product)}$$

Find the molecular weight of the major product R

43. 2.5g of a mixture containing $CaCO_3$, $Ca(HCO_3)_2$ and $NaCl$ was dissolved in 100 ml water and its 10 ml portion required 10 ml 0.05M H_2SO_4 solution to reach the phenolphthalein end point. An another 10 ml portion of the same stock solution required 32.35 ml of the same acid solution to reach the methyl orange end point. Determine mass percentage of $NaCl$ in the original mixture.

44. Some amount of Al(s) is taken with $Cl_2(g)$ in a cylinder fitted with piston at 300 K and 1 bar pressure. Now mixture is heated and Al reacts with chlorine with a spectacular display of spark.

$$2Al(s)+3Cl_2(g)\longrightarrow 2AlCl_3(s) \qquad \Delta H^o = 342\,K\,cal\,mol^{-1}$$

Given, area of piston is $500\,cm^2$ and it moves by 48 cm. Calculate heat change at constant pressure in kCal for the above process. Use: $R = 0.08\,L\,barK^{-1}\,mol^{-1}$

45. In an unielectronic species of unknown atomic number a transition of electron from a higher energy level to ground state resulted in emission of a photon having energy approximately 108.9 eV. If the change in wavelength of the electron in the two levels is two times the circumference of its first Bohr orbit then what would be the atomic number (Z) of the unieletronic species.

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46. Calculate density $(gm/litre)$ of a real gas at 2 atm pressure (low) and 500K if molar mass = 164.2 gm/mole and Vander Waal constants a and b are

$a = 4.105\, atm - litre^2 / mole^2$

and $b = 0.1\, litre/mole$.

[$R = 0.0821$ atm-litre/mole kelvin]

47. Consider the following reaction:

$$MnS + 2HCl \longrightarrow X + Y$$

$$X + 2NaOH \longrightarrow \underset{(white\ ppt)}{Q} + \text{other product}$$

$$Q + [O] \xrightarrow{(atmospheric\ oxidation)} \underset{(Brown\ colour)}{Z}$$

If the reaction is started with 87 grams of MnS, the weight (in grams) of brown coloured compound (Z) obtained is____.

(Given that : GAW of Mn=55,S=32, H=1,Cl=35.5)

48. In one of the experiment, $NiCl_2$ is dissolved in water then a complex 'X' is formed. If the bidentate ligand, ethane-1,2-diamine is progressively added to the 'X' in the molar ratios $en : Ni$, 1:1, 2:1, 3:1; three complexes (P,Q,R) are obtained. Then the total number of isomers of Q and R complexes is_____. (ignore hydrate isomers)

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31	95	10	98	579

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29	113	205	745	4,212

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